Finding 4 - Snort Setup and Starter Rules (Ubuntu VM)

Objective

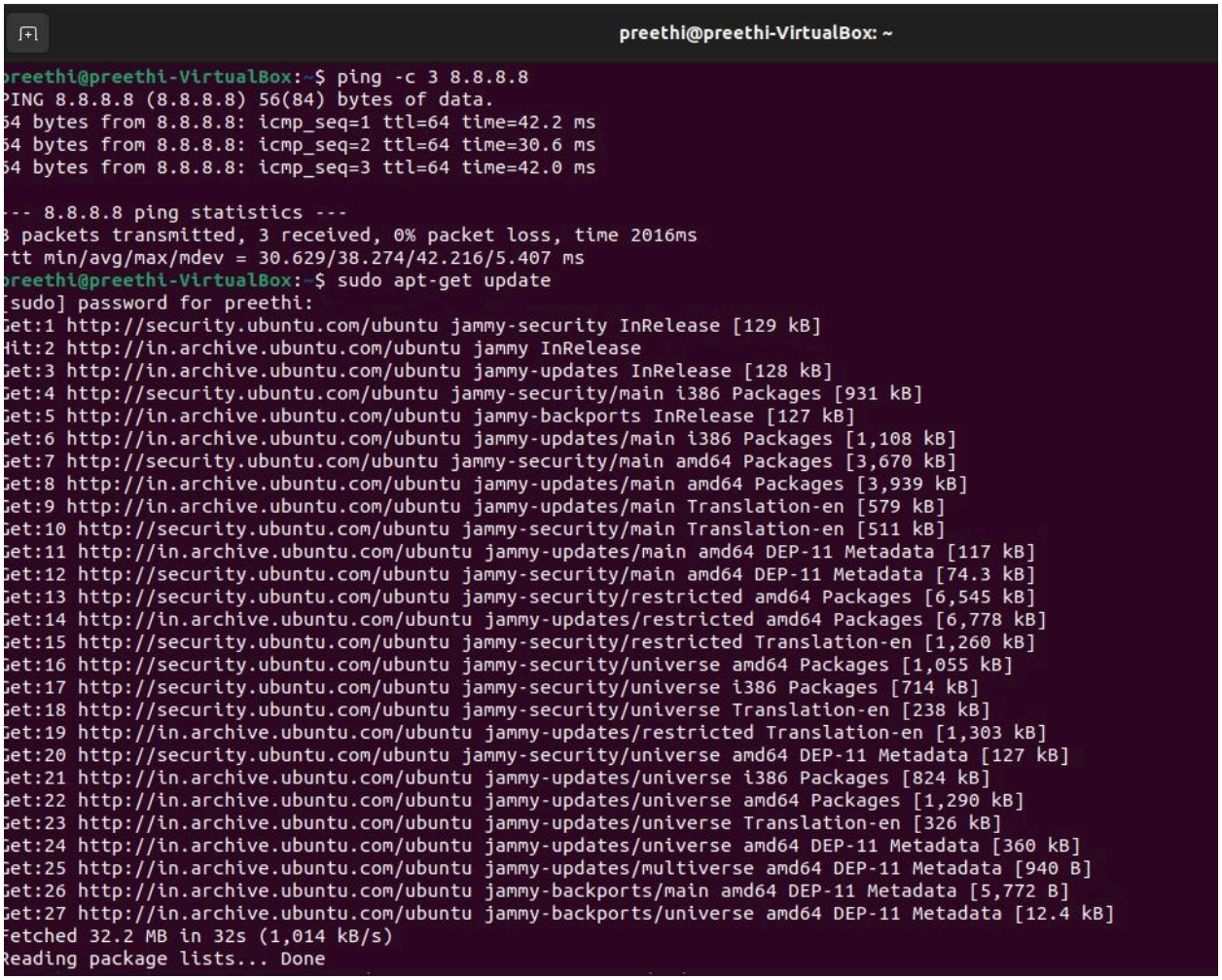
Install Snort, configure `HOME_NET` and `EXTERNAL_NET`, add the starter rules and confirm they generate alerts.

Step 0 - Identify the network interface

VirtualBox Ubuntu guests usually use `enp0s3` (or similar), not `eth0`.

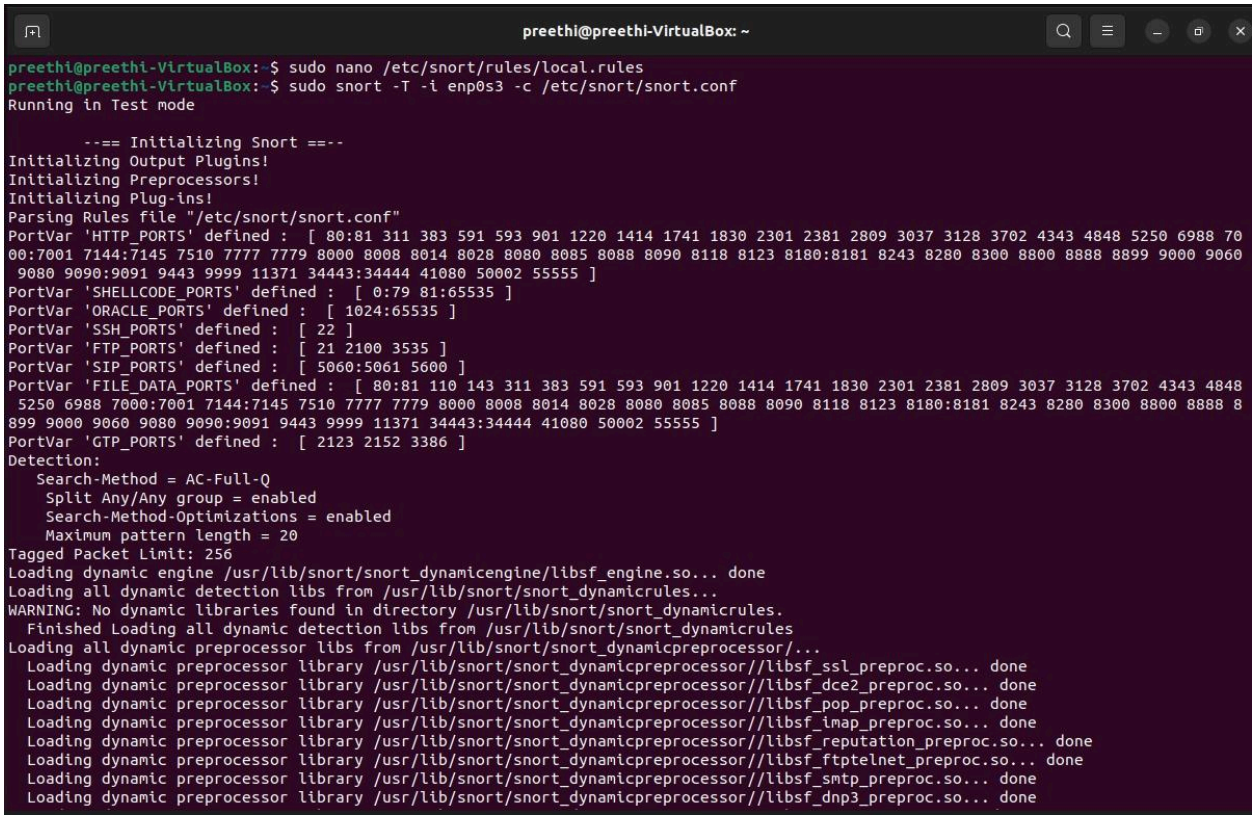
```
ip a
```

Interface used in this lab: the name shown in Screenshot 4.1 (replace `<interface>` in all commands below with that name).



Installation

Step	Command
1. Update package repository	`sudo apt-get update`
2. Install Snort and default rules	`sudo apt-get install -y snort snort-common snort-rules-default`
3. Verify installation	`snort -V`

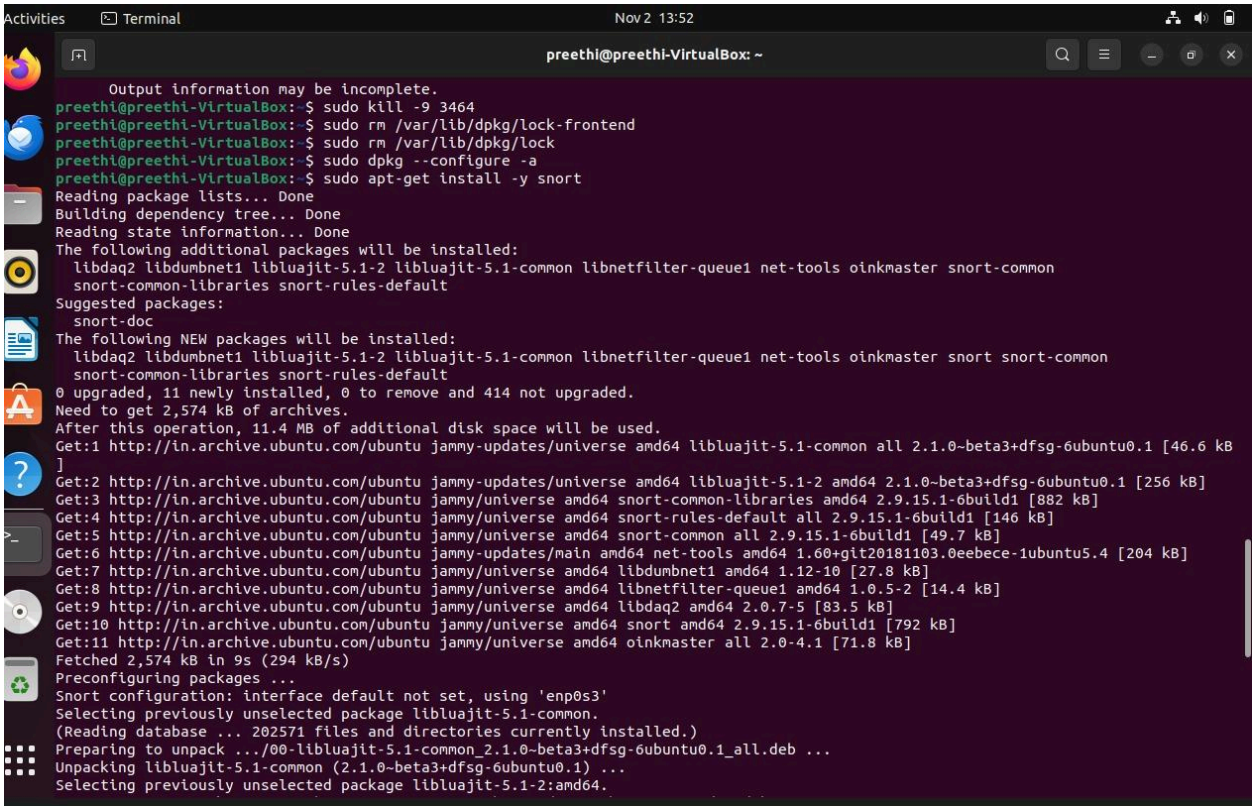


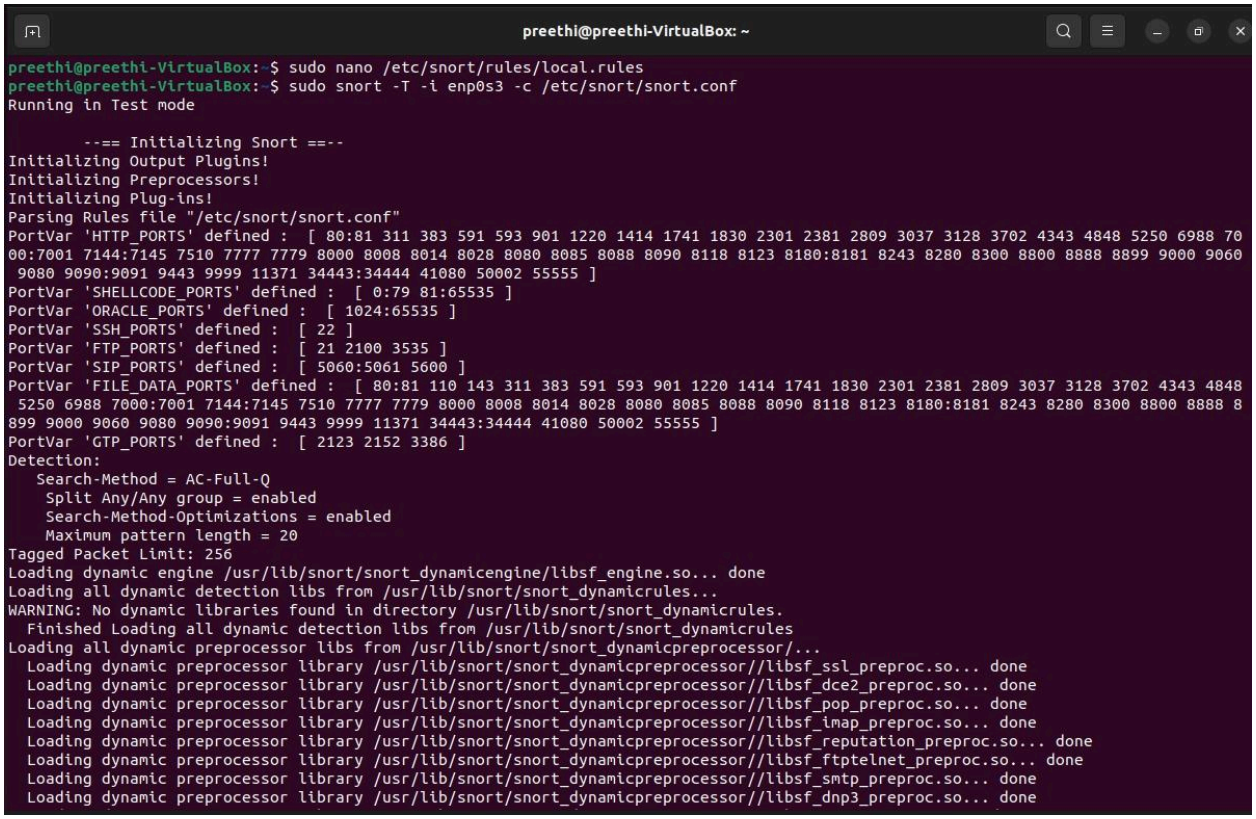
Configuration

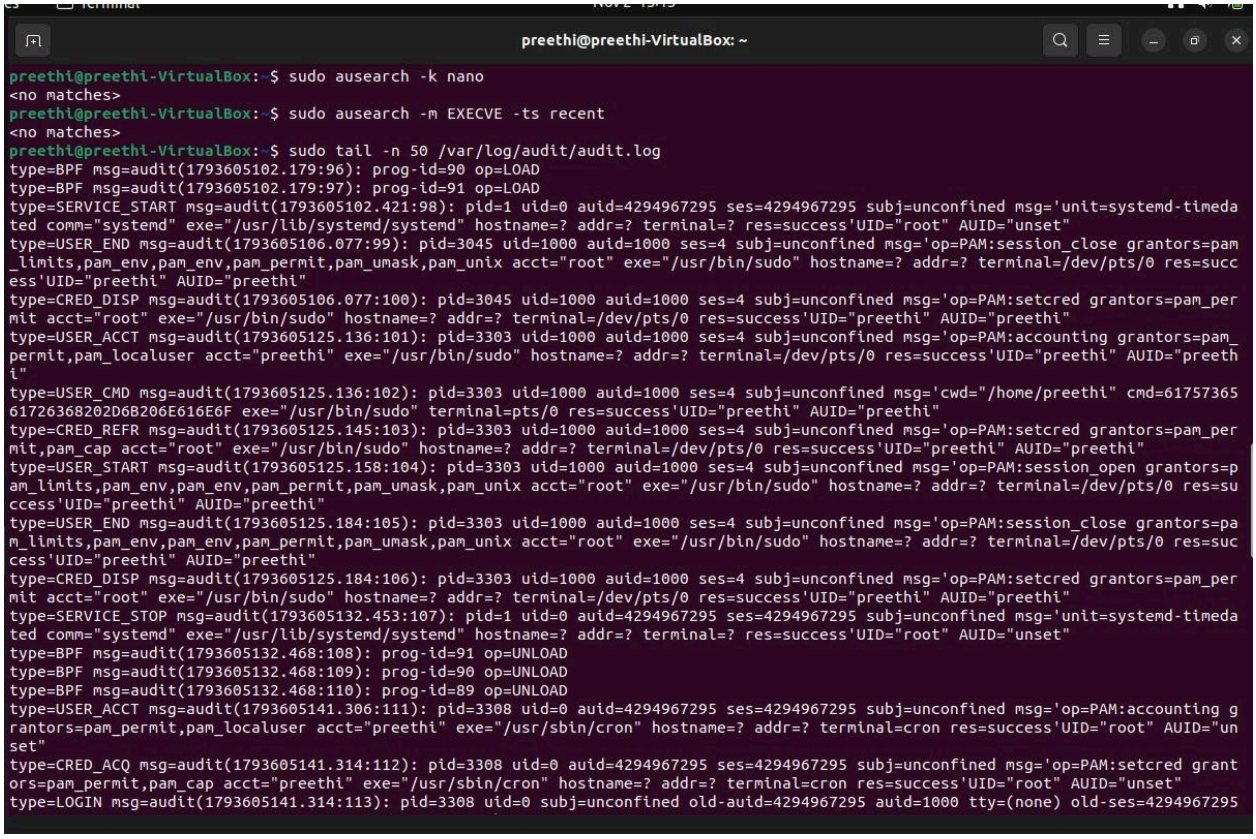
Edit the configuration file:

```
sudo nano /etc/snort/snort.conf
```

Variable	Value set
`HOME_NET`	The local subnet of the VM, for example `192.168.1.0/24` (value as shown in Screenshot 4.3)
`EXTERNAL_NET`	`!$HOME_NET`


Activities
Terminal
Nov 2 13:52
preethi@preethi-VirtualBox: ~
Output information may be incomplete.
preethi@preethi-VirtualBox:~$ sudo kill -9 3464
preethi@preethi-VirtualBox:~$ sudo rm /var/lib/dpkg/lock-frontend
preethi@preethi-VirtualBox:~$ sudo rm /var/lib/dpkg/lock
preethi@preethi-VirtualBox:~$ sudo dpkg --configure -a
preethi@preethi-VirtualBox:~$ sudo apt-get install -y snort
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
libdaq2 libdumbnet1 libluajit-5.1-2 libluajit-5.1-common libnetfilter-queue1 net-tools oinkmaster snort-common
snort-common-libraries snort-rules-default
Suggested packages:
snort-doc
The following NEW packages will be installed:
libdaq2 libdumbnet1 libluajit-5.1-2 libluajit-5.1-common libnetfilter-queue1 net-tools oinkmaster snort snort-common
snort-common-libraries snort-rules-default
0 upgraded, 11 newly installed, 0 to remove and 414 not upgraded.
Need to get 2,574 kB of archives.
After this operation, 11.4 MB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 libluajit-5.1-common all 2.1.0~beta3+dfsg-6ubuntu0.1 [46.6 kB
]
Get:2 http://in.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 libluajit-5.1-2 amd64 2.1.0~beta3+dfsg-6ubuntu0.1 [256 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 snort-common-libraries amd64 2.9.15.1-6build1 [882 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 snort-rules-default all 2.9.15.1-6build1 [146 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 snort-common all 2.9.15.1-6build1 [49.7 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 net-tools amd64 1.60+git20181103.0eebece-1ubuntu5.4 [204 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 libdumbnet1 amd64 1.12-10 [27.8 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 libnetfilter-queue1 amd64 1.0.5-2 [14.4 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 libdaq2 amd64 2.0.7-5 [83.5 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 snort amd64 2.9.15.1-6build1 [792 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu jammy/universe amd64 oinkmaster all 2.0-4.1 [71.8 kB]
Fetched 2,574 kB in 9s (294 kB/s)
Preconfiguring packages ...
Snort configuration: interface default not set, using 'enp0s3'
Selecting previously unselected package libluajit-5.1-common.
(Reading database ... 202571 files and directories currently installed.)
Preparing to unpack .../00-libluajit-5.1-common_2.1.0~beta3+dfsg-6ubuntu0.1_all.deb ...
Unpacking libluajit-5.1-common (2.1.0~beta3+dfsg-6ubuntu0.1) ...
Selecting previously unselected package libluajit-5.1-2:amd64.


preethi@preethi-VirtualBox: ~
preethi@preethi-VirtualBox:~$ sudo nano /etc/snort/rules/local.rules
preethi@preethi-VirtualBox:~$ sudo snort -T -i enp0s3 -c /etc/snort/snort.conf
Running in Test mode
--== Initializing Snort ==--
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "/etc/snort/snort.conf"
PortVar 'HTTP_PORTS' defined : [80:81 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848 5250 6988 70
00:7001 7144:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8899 9000 9060
9080 9090:9091 9443 9999 11371 34443:34444 41080 50002 55555]
PortVar 'SHELLCODE_PORTS' defined : [0:79 81:65535]
PortVar 'ORACLE_PORTS' defined : [1024:65535]
PortVar 'SSH_PORTS' defined : [22]
PortVar 'FTP_PORTS' defined : [21 2100 3535]
PortVar 'SIP_PORTS' defined : [5060:5061 5600]
PortVar 'FILE_DATA_PORTS' defined : [80:81 110 143 311 383 591 593 901 1220 1414 1741 1830 2301 2381 2809 3037 3128 3702 4343 4848
5250 6988 7000:7001 7144:7145 7510 7777 7779 8000 8008 8014 8028 8080 8085 8088 8090 8118 8123 8180:8181 8243 8280 8300 8800 8888 8
899 9000 9060 9080 9090:9091 9443 9999 11371 34443:34444 41080 50002 55555]
PortVar 'GTP_PORTS' defined : [2123 2152 3386]
Detection:
Search-Method = AC-Full-Q
Split Any/Any group = enabled
Search-Method-Optimizations = enabled
Maximum pattern length = 20
Tagged Packet Limit: 256
Loading dynamic engine /usr/lib/snort/snort_dynamicengine/libsf_engine.so... done
Loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules...
WARNING: No dynamic libraries found in directory /usr/lib/snort/snort_dynamicrules.
Finished Loading all dynamic detection libs from /usr/lib/snort/snort_dynamicrules
Loading all dynamic preprocessor libs from /usr/lib/snort/snort_dynamicpreprocessor/...
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsf_ssl_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsf_dce2_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsf_pop_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsf_imap_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsf_reputation_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsf_ftptelnet_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsf_smtp_preproc.so... done
Loading dynamic preprocessor library /usr/lib/snort/snort_dynamicpreprocessor//libsf_dnp3_preproc.so... done



Starter rules

Open the local rules file:

```
sudo nano /etc/snort/rules/local.rules
```

Rules added:

```
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping Detected"; sid:1000001;
rev:1;)
alert tcp any any -> $HOME_NET 22 (msg:"SSH Connection Attempt"; sid:1000002;
rev:1;)
alert tcp any any -> $HOME_NET 80 (msg:"HTTP Traffic"; sid:1000003; rev:1;)
```

```
  GNU nano 6.2                                    /etc/snort/rules/local.rules
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# ----------------
# LOCAL RULES
# ----------------
# This file intentionally does not come with signatures.  Put your local
# additions here.
# alert icmp any any -> $HOME_NET any (msg:"ICMP ping detected"; sid:1000001; rev:1;)
# alert tcp any any -> $HOME_NET any (msg:"SSH connection attempt"; sid:1000002; rev:1;)
# altert tcp any any -> $HOME_NET any ("HTTP traffic"; sid:1000003; rev:1;)
```

Rule explanation

SID	Rule	What it detects
1000001	ICMP Ping Detected	Any ICMP packet to the home network (ping sweeps, reconnaissance)
1000002	SSH Connection Attempt	Any TCP packet to port 22 (SSH access monitoring)
1000003	HTTP Traffic	Any TCP packet to port 80 (web traffic monitoring)

Test the configuration

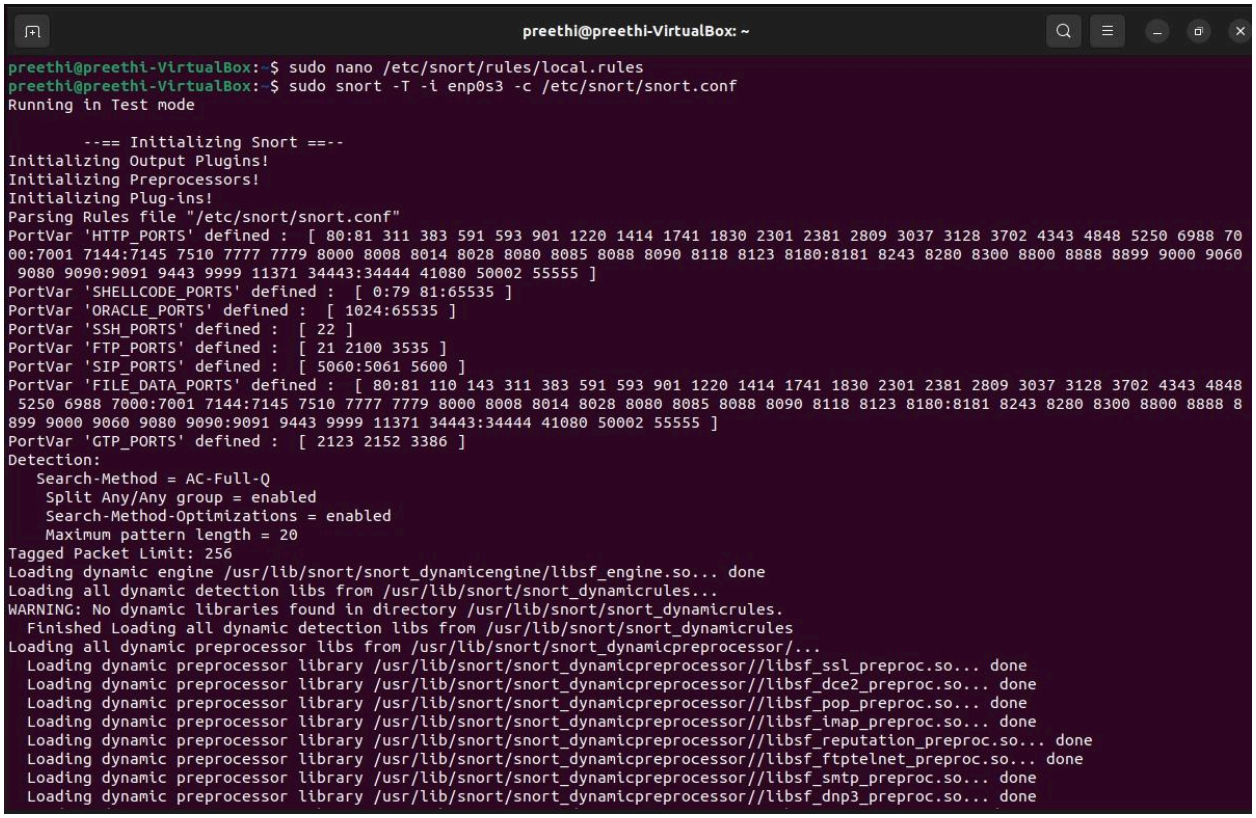
```
sudo snort -T -i <interface> -c /etc/snort/snort.conf
```

A successful test ends with the message "Snort successfully validated the configuration!"

`[Insert Screenshot 4.5 here - successful configuration validation]`

Start Snort in console mode

```
sudo snort -A console -q -c /etc/snort/snort.conf -i <interface>
```



Notes

- `-A console` prints alerts to the terminal, `-q` removes the startup banner, `-c` selects the config file and `-i` selects the interface.
- Make sure `include $RULE_PATH/local.rules` is enabled in `snort.conf`.
- Keep each `sid` unique (1,000,000 and above for local rules).

SOC relevance

Starter rules give immediate visibility of reconnaissance (ping), remote administration attempts (SSH) and web traffic. They also teach the rule workflow used in every IDS deployment: write, validate, run, test, tune.